# ARIAN RAHMAN ADITTA

**Electronics & Communication Engineering Graduate | RF/mmWave IC Design • Analog/RF IC • VLSI/EDA**

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**Academic Website** | **LinkedIn** | **Google Scholar** | **GitHub**

## RESEARCH INTERESTS

• RF/mmWave IC Design • Analog/RF IC Design • RFIC Design Automation • ML-Assisted Hardware Design
• SiGe BiCMOS Front-End Circuits

## EDUCATION

**B.Sc. Engineering in Electronics and Communication Engineering** *[2022 - 2026]*

*Khulna University of Engineering and Technology, Khulna, Bangladesh*

**CGPA:** 3.71/4.00 | **Class Rank:** 5/60 **(Top 8.3%)**

**Final four term GPAs:** *3.89, 4.00, 4.00, 3.88*

**Selected Coursework:** VLSI Design & Nanotechnology, Analog Electronics, Microwave Engineering, Digital Communications, Antenna Engineering, Signals & Systems, Digital Signal Processing, Microprocessors & Microcomputers, Digital Electronics & Logic Circuits, Machine Learning

## RESEARCH EXPERIENCE

### Open-Source RFIC Design and Verification of a 28-GHz Two-Stage Class-A SiGe Power Amplifier with Machine-Learning-Assisted Optimization

***Undergraduate Thesis, Department of ECE, KUET | Supervisor: Aysha Tabassum | 2026 | Ongoing***

- Designed and simulated a **two-stage 28-GHz Class-A SiGe PA** using the **IHP SG13G2 130-nm SiGe BiCMOS Open PDK**, with an open-source RFIC workflow spanning **Qucs-S, ngspice/Xyce, KLayout** and **EM-aware** modeling.
- Developed a progressive **ideal-passive → PDK-aware → EM-aware** realization flow, with the final process-aware configuration achieving **17.4 dB gain at 28 GHz** while quantifying the impact of physical passive realization.
- Used **machine learning** as a supporting design-space exploration layer while retaining circuit simulation and RF verification as the primary evidence; currently extending the work toward **layout/GDSII-level** realization, with a **first-author** thesis-derived manuscript submitted to **ICCIT 2026**.

### 27-GHz SiGe RF Front-End Design-Space Generation

***Mentored Research with Sultan Muhammad Ikram, RF/mmWave MMIC Designer, Neural Semiconductor Limited | 2026–Present | GitHub***

- Designed parameterized **common-emitter and cascode SiGe PA topologies**, systematically varying device sizing, bias conditions, RF chokes and input/output matching networks within the **IHP SG13G2** platform.
- Building a **private circuit parameter–performance dataset** through repeatable DC, small-signal, stability and large-signal characterization using **Qucs-S, ngspice, Xyce and Python**-based post-processing.
- Extending the same framework toward **common-emitter and cascode LNAs**, creating a **unified PA/LNA** design space for comparative topology analysis, surrogate modeling and future **ML-assisted** RFIC design exploration.

### GNN-SABLE: Learning-Guided Constraint-Rich VLSI Floorplanning

***Independent Research | 2026–Present | First-author manuscript submitted to ICEEICT 2027 | Ongoing***

- Developed a ***GNN-guided B*-tree floorplanning framework*** combining learned initialization with simulated annealing, legality-verified refinement and constraint-aware repacking while keeping geometric legality algorithmically enforced.
- Evaluated the complete pipeline on an independent **100-layout confirmation panel with 100/100 hard-legal outputs**, then used controlled ablations to distinguish learned guidance from the underlying classical optimization pipeline.

### Source-Aware RTL-to-QoR Prediction for Early Area and Timing Estimation

***Research Collaboration with Zakia Sultana, Ph.D. Candidate, MSCAD Laboratory, University of Arkansas | Ongoing | Manuscript in Preparation | GitHub***

- Developed a **source-aware RTL-to-QoR framework** for **mapped-area** and **standardized pre-layout timing** prediction using structural RTL representations, source-connected grouping and independently generated **Yosys/ABC/OpenSTA** physical labels.
- Built the evaluation around **681 labelled configurations grouped into 498 source components**, with model-specific prediction freezes and fresh prospective cohorts; the study includes robustness, runtime and reproducibility analyses supporting the journal manuscript currently in preparation.

## PUBLICATIONS & MANUSCRIPTS

### Peer-Reviewed Publications

[1] **A. R. Aditta**, P. K. Das, U. R. Arshee, and A. B. M. Aowlad Hossain, "SenseStick: A Bangla-English Edge-AI Cane for Road-Safety and Biomedical Reliability Monitoring," *IEEE BECITHCON, 2026.* **[First Author]**

[2] P. K. Das, S. M. R. Islam, and **A. R. Aditta**, "LMatchNet-V: Physics-Seeded Learning for Vendor-Aware RF L-Section Matching with Adaptive Safe-Library Search," *IEEE MARSCON, 2026.*

[3] M. A. Jahin, S. Soudeep, **A. R. Aditta**, M. F. Mridha, N. Fahad, and M. J. Hossen, "Vision Transformers for End-to-End Quark-Gluon Jet Classification from Calorimeter Images," *Communications in Computer and Information Science*, vol. 2640, Springer, pp. 135–150, 2025. DOI: 10.1007/978-981-95-0988-1_10

### Submitted Manuscripts

[4] **A. R. Aditta** and A. Tabassum, "Progressive PDK-to-EM Realization of a 28 GHz Two-Stage Class-A SiGe Power Amplifier in an Open-Source RFIC Flow," submitted to the *29th International Conference on Computer and Information Technology (ICCIT 2026)*. **[First Author]**

[5] **A. R. Aditta**, M. M. Zaman, M. S. U. Alam, and S. M. R. Islam, "Graph Neural Network Guidance for B*-Tree Annealing in Constraint-Rich Outline-Aware Floorplanning," submitted to the *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2027)*. **[First Author]**

[6] "Source-Aware Evaluation of RTL Area and Timing Prediction with Prospective Confirmation." Manuscript in preparation.

## TECHNICAL SKILLS

- **RFIC / Circuit / EM Tools:** Qucs-S, ngspice, Xyce, KLayout, openEMS, CST Studio Suite, LTspice, Cadence Virtuoso, IHP SG13G2 Open PDK
- **VLSI / EDA / RTL Tools:** Verilog, Yosys, OpenSTA
- **Programming & Scientific Computing:** Python, C, C++, MATLAB, NumPy, Pandas, Matplotlib
- **Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, GNNs, CNNs, tree-based models, regression/classification, model evaluation
- **Embedded Systems:** Raspberry Pi, ESP32, Arduino, OpenCV
- **Research Tools:** Git, LaTeX, Zotero

## HONORS & AWARDS

**Dean's Award, 3rd & 4th** Year, Khulna University of Engineering & Technology
- Awarded for outstanding academic performance in consecutive senior academic years.

**Technical Scholarship,** Khulna University of Engineering and Technology (KUET)
- Awarded for academic performance during the first three undergraduate years.

## ACADEMIC & RESEARCH LEADERSHIP

**President, KUET Research Society, ECE** *[2025-2026]*
- Organized a department-wide pre-mid thesis competition involving 3rd- and 4th-year ECE students to promote research presentation, peer learning and early thesis engagement.
- Coordinated online graduate-study and research guidance sessions with KUET alumni pursuing M.S./Ph.D. degrees abroad, helping current students understand research preparation, graduate applications, and academic career pathways.

**General Secretary, KUET Debating Society** *[2023-2026]*
- Led organizational activities and competitive debating initiatives, developing extensive experience in communication, team leadership, public speaking, team coordination and event management.

## RELEVANT PROFESSIONAL EXPERIENCE

**Campus Ambassador, Ulkasemi** *[Aug 2024 – Sep 2025]*
- Supported KUET–Ulkasemi semiconductor-industry engagement and student outreach; helped organize a five-day career and management training program for final-year students.

**Data Analyst Intern, WEGRO** *[Mar 2025 – Jul 2025]*
- Analyzed business data, created decision-oriented visualizations, and contributed to data-driven process improvement in an agri-fintech startup.